

# Privacy-Preserving Visualization of Brain Functional Connectivity

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## Abstract

Visualizations are fundamental to neuroimaging research, facilitating tasks ranging from exploratory data analysis to the communication and interpretation of findings. Despite their necessity, visualizations can potentially compromise the confidentiality of individual participants. In this paper, we discuss how visualizations may inadvertently lead to privacy leakage and explore methods to mitigate such risks. Our work investigates ways to securely share visualizations that faithfully preserve the patterns supporting the derived insights from data analysis, rather than deriving conclusions from the visualizations themselves. We address privacy-preserving visualization within a differential privacy (DP) framework, focusing on commonly used visualization methods for functional network connectivity. Various perturbation-based strategies are investigated for protecting correlation-related measures, with analyses of their privacy costs and the effects of pre-processing and/or post-processing. To achieve a better balance between privacy and visual utility, this paper introduces novel workflows tailored for connectogram and seed-based connectivity visualizations that maintain the qualitative patterns observed in non-private results. Overall, this work illustrates how DP can be effectively applied to neuroimaging visualization, demonstrating its efficacy as a robust methodology for securing sensitive biomedical data.

*Keywords:*

## 1. Introduction

Data visualization (Kostelnick, 2008; Bhattacharjee et al., 2020) is an integral part of the research workflow, serving distinct purposes at different stages of analysis. In the early phase of exploratory data analysis, it enables researchers to examine data distributions and uncover underlying patterns. At later stages, visualizations facilitate the communication of analytical findings and aid in the interpretation of results. While privacy concerns are often discussed in the context of sharing raw data or summary statistics (Sweeney, 1997; Narayanan and Shmatikov, 2008), visualizations derived from these data can also expose sensitive information, particularly in domains involving human health and behavior, yet this aspect has received comparatively less attention than data leakage itself.

A range of privacy attacks have demonstrated that anonymized or aggregated outputs can still lead to re-identification or data leakage. Classic examples include linking de-identified health records with publicly available information (Sweeney, 2002) and uncovering user identities from anonymized movie rating data (Narayanan and Shmatikov, 2008). Similar risks extend to visualizations, where adversaries may exploit graphical cues and/or external sources to infer sensitive information. A well-known real-world example is Strava’s release of an anonymized global activity heatmap, which unintentionally exposed secret U.S. military bases (Allen, 2020). Comparable risks arise in scientific visualizations. For example, scatterplots or boxplots commonly used in neuroimaging exploratory data analysis may reveal actual data values. Likewise, when a research team publishes average fMRI functional network connectivity (FNC) matrices for healthy and diseased groups, re-identification becomes possible: if an attacker knows the FNC of all subjects except one, the group membership of the remaining individual can be inferred by comparing the published visualizations (see Figure 1). Although this represents an extreme case, it illustrates how visualization can lead to privacy leakage, as auxiliary information about individuals may be accessible through various external sources or linkage attacks. This example also motivates the need for privacy-preserving visualization methods that remain robust even when an attacker possesses substantial background knowledge. By contrast, even when the attacker has only limited knowledge, published

visualizations can still enable inference if privacy is not explicitly considered. Concretely, one can adapt membership inference tests (Homer et al., 2008; Dwork et al., 2015) to the visualization setting. Representing a published visualization as  $v_{\text{pub}}$  generated from a group, a target individual’s visualization as  $v_{\text{target}}$ , and a reference sample as  $v_{\text{ref}}$ , an adversary can compare similarity scores such as  $\langle v_{\text{target}}, v_{\text{pub}} \rangle$  and  $\langle v_{\text{ref}}, v_{\text{pub}} \rangle$ , and determine whether the target belongs to the group based on their difference. Although such tests are abstract and rely on modeling assumptions, they demonstrate that releasing aggregate or visual summaries can still leak membership information in the absence of explicit privacy protection.

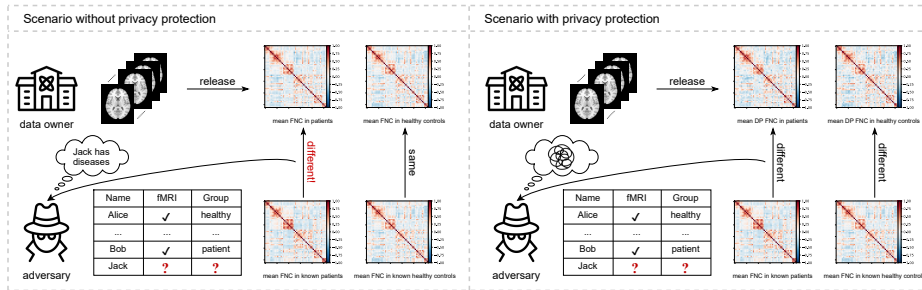


Figure 1: In a scenario without privacy protection, an adversary can infer information about unknown individuals. However, with differential privacy, the adversary cannot make such inferences.

Methods such as differential privacy (DP) (Dwork, 2006) offer protection against data leakage risks by introducing carefully calibrated randomness into the released outputs, thereby preventing accurate inference about any individual even when an adversary possesses substantial background knowledge (see Section 2.1 for details). This protection often comes at the cost of a loss of accuracy and utility, which raises an important question: can privacy-preserving visualizations still capture the same patterns and convey the intended information? Previous study on differentially private visualizations, including boxplots, violin plots, and scatterplots (Tao and Sarwate, 2025; Panavas et al., 2023), has demonstrated that these methods can effectively protect individual information while preserving the essential patterns intended to be communicated (see Figure 2). Motivated by the growing need for privacy protection and the previous findings, this paper examines how privacy considerations can be incorporated into neuroimaging visualizations. Specifically, we explore the application of DP to commonly used

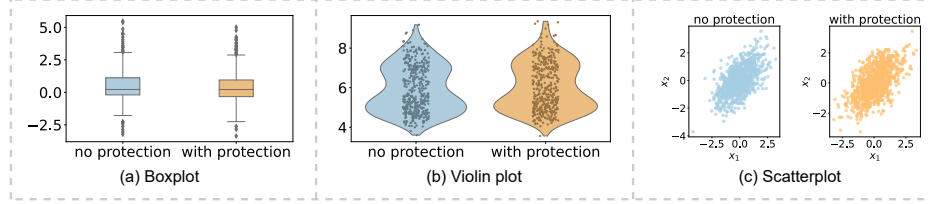


Figure 2: Comparison of visualizations with and without DP protection. DP aims to protect individual-level information, such as demographic or clinical measurements, while preserving overall statistical patterns in the data, enabling meaningful interpretation of population-level structure without relying on exact individual values. In the unprotected setting, visualizations are generated directly from raw individual records. In contrast, under DP, individual-level information is protected while overall distributional patterns are preserved; for example, boxplots show distribution and variability, violin plots reveal multimodal features, and scatterplots illustrate trends between variables. The displayed points no longer represent original individual data values and instead serve only to convey underlying characteristics.

visualizations, including FNC, connectogram, and seed-based connectivity (SBC) plots (Van Den Heuvel and Pol, 2010; Mohanty et al., 2020; Joel et al., 2011; Chopra et al., 2023). Our main contributions are summarized as follows.

- We propose a series of novel methods for DP visualization in neuroimaging, including pre- and post-processing strategies to enhance visual quality.
- We systematically evaluate these methods across multiple real-world datasets, comparing the performance of different mechanisms and identifying their optimal composition strategies under varying privacy parameters.
- We develop end-to-end differentially private workflows for connectogram and SBC visualizations, illustrating group comparisons and significant connectivities.
- We assess the interpretability and utility of the proposed visualizations through both qualitative and quantitative analyses.



## 2. Background

### 2.1. Differential Privacy

As data collection, storage, and analysis capabilities have improved significantly, privacy risks have increased, driving a growing focus on privacy research over the past two decades. Earlier syntactic approaches to privacy, such as de-identification and  $k$ -Anonymity (Garfinkel, 2015; Sweeney, 2002), are susceptible to linkage or membership inference attacks (Dwork et al., 2017; Merener, 2012; Ganta et al., 2008; Henriksen-Bulmer and Jeary, 2016; Fiore et al., 2020). Unlike these approaches, DP (Dwork et al., 2006; Dwork and Roth, 2014) has emerged as a state-of-the-art approach that provides a quantitative framework for privacy protection, addressing the vulnerabilities of earlier syntactic methods. It enables statistical analyses on datasets while ensuring that the privacy of individuals whose records are included is preserved. A differentially private algorithm or mechanism is a randomized function: the randomization is designed to make it difficult for an adversary observing the output to discern private information. The kind of information that DP protects is captured by the notion of neighboring databases. For two datasets  $D$  and  $D'$  that are neighbors, denoted as  $D \sim D'$ , the randomization should make it difficult for the adversary to distinguish, using a hypothesis test, whether the output was generated from  $D$  or  $D'$  (Wasserman and Zhou, 2010; Kairouz et al., 2015). Most commonly, we say  $D \sim D'$  if they differ by the data of a single individual: this means that an adversary cannot reliably tell whether that individual was in the dataset or not. A full description of the key notations used in the paper is provided in Table 1. Some minor abuse of notation may occur, but it should be clear from the context.

For an algorithm to be differentially private, it must guarantee that the adversary’s test is unreliable for all possible  $D \sim D'$ . This unreliability is measured in DP using two parameters,  $\epsilon$  and  $\delta$  (see Definition 1). The parameter  $\epsilon$  denotes the privacy budget and controls the maximum divergence between the output distributions of an algorithm  $M$  evaluated on any pair of neighboring datasets. Lower values of  $\epsilon$  lead to stronger privacy guarantees but typically result in decreased accuracy. The parameter  $\delta$  quantifies the probability of a small relaxation in the differential privacy guarantee. In practice,  $\delta$  is typically chosen to be much smaller than the inverse of the dataset size, i.e.,  $\delta \ll 1/N$ . The definition of DP ensures that, for any pair of neighboring datasets  $D$  and  $D'$ , the privacy loss (Dwork and Roth, 2014) exceeds  $\epsilon$  with probability at most  $\delta$ .

| Notation                              | Description                                                       |
|---------------------------------------|-------------------------------------------------------------------|
| $\epsilon$                            | Privacy budget                                                    |
| $\delta$                              | Probability of privacy violation or failure                       |
| $M$                                   | Randomized algorithm or mechanism                                 |
| $D \sim D'$                           | Neighboring datasets                                              |
| $\Delta_1, \Delta_2$                  | Global $\ell_1$ -sensitivity and $\ell_2$ -sensitivity of $f(D)$  |
| $\Delta_u$                            | Sensitivity of the utility function $u$                           |
| $v$                                   | Brain voxel                                                       |
| $r$                                   | A single ROI or $r$ brain regions, as defined by context          |
| $t, T$                                | Timepoint and the total number of timepoints, respectively        |
| $S(\cdot, t)$                         | BOLD value for a voxel or an ROI at time $t$                      |
| $X_{ij}$                              | Correlation coefficient between the timeseries of $r_i$ and $r_j$ |
| $X^{(n)} \in \mathbb{R}^{r \times r}$ | FNC matrix for subject $n$                                        |
| $\bar{X}, \tilde{X}$                  | Average FNC and perturbed FNC                                     |

Table 1: Key notations used in the paper.

**Definition 1** ( $(\epsilon, \delta)$ -DP (Dwork and Roth, 2014)). Let  $\epsilon, \delta \geq 0$ , a randomized algorithm  $M : \mathcal{D} \rightarrow \mathcal{H}$  is  $(\epsilon, \delta)$ -differentially private if for all  $\mathcal{Y} \subseteq \mathcal{H}$  and for all neighboring datasets  $D, D' \in \mathcal{D}$ ,

$$\mathbb{P}[M(D) \in \mathcal{Y}] \leq \exp(\epsilon) \mathbb{P}[M(D') \in \mathcal{Y}] + \delta.$$

A common approach to designing an  $(\epsilon, \delta)$ -DP algorithm for approximating a function  $f : \mathcal{D} \rightarrow \mathbb{R}^d$  is to add appropriately calibrated random noise to the output  $f(D)$ . The required noise level depends not only on  $\epsilon$  and  $\delta$ , but also on the sensitivity of the function, defined as the maximum change in the output induced by modifying a single data point. We define the global  $\ell_1$  and  $\ell_2$  sensitivities of  $f$  as

$$\Delta_i = \max_{D \sim D'} \|f(D) - f(D')\|_i$$

for  $i = 1$  or  $2$ , where  $\|\cdot\|_i$  denotes the  $\ell_i$ -norm. “Pure”  $(\epsilon, 0)$ -DP is achieved via the Laplace mechanism (Dwork and Roth, 2014), which perturbs the output of  $f(D)$  with Laplace noise. Formally,  $M_L(D) = f(D) + Z$ , where  $Z \sim \text{Laplace}(0, \Delta_1/\epsilon \cdot I)$ . The identity matrix  $I \in \mathbb{R}^d$  is included when  $f(D)$  is high-dimensional. Alternative noise distributions have also been proposed (Geng et al., 2015; Alghamdi et al., 2022b). The Gaussian mechanism provides “approximate”  $(\epsilon, \delta)$ -DP by adding Gaussian noise to the output of  $f$ ,

i.e.,  $M_G(D) = f(D) + Z$ , where  $Z \sim \mathcal{N}(0, \sigma^2 \cdot I)$  and  $\sigma = \Delta_2 \sqrt{2 \log(1.25/\delta)}/\epsilon$ . However, this standard choice of  $\sigma$  is known to be overly conservative in the high-privacy regime ( $\epsilon \rightarrow 0$ ), and may not be appropriate in low-privacy settings ( $\epsilon \rightarrow \infty$ ). To address this, [Balle and Wang \(2018\)](#) proposed an “analytic” formula for the variance. In the following sections and experiments, we use the analytic Gaussian mechanism rather than the classical Gaussian mechanism when considering noise from the Gaussian distribution.

Additive noise mechanisms are well-suited for privatizing real-valued functions. In other scenarios, algorithms may instead randomly select an output from a set of choices. In such cases, the exponential mechanism ([McSherry and Talwar, 2007](#)) enables  $(\epsilon, 0)$ -DP randomized selection from a discrete set. Sometimes, numerical outputs can be mapped to a discrete set. For example, in visualization, certain numerical values are mapped to the same color. The exponential mechanism operates over an output space  $\mathcal{H}$  using a utility function  $u : \mathcal{D} \times \mathcal{H} \rightarrow \mathbb{R}$  that assigns a score to each input-output pair. Given a dataset  $D$ , it samples an element  $H \in \mathcal{H}$  with probability proportional to  $\exp\left(\frac{\epsilon u(D, H)}{2\Delta_u}\right)$ , where  $\Delta_u = \max_{H \in \mathcal{H}} \max_{D \sim D'} |u(D, H) - u(D', H)|$  denotes the sensitivity of the utility function.

Mechanisms such as additive noise or randomized selection protect privacy in a single analysis. When applied multiple times, the accumulated privacy loss can be formally quantified using the composition property of DP. There are several types of composition rules: basic, advanced, exact, and Rényi differential privacy (RDP)-based compositions ([Dwork et al., 2010](#); [Dwork and Roth, 2014](#); [Kairouz et al., 2015](#); [Murtagh and Vadhan, 2015](#); [Abadi et al., 2016](#); [Mironov, 2017](#); [Balle et al., 2020](#); [Dong et al., 2022](#); [Sommer et al., 2018](#); [Alghamdi et al., 2022b](#); [a](#); [Koskela and Honkela, 2020](#); [Gopi et al., 2021](#); [Koskela et al., 2020](#); [Ghazi et al., 2022](#); [Doroshenko et al., 2022](#)). A full discussion of composition rules is beyond the scope of this paper, but for the purposes of this work, advances in composition analyses can only improve the guarantees that we make. Another important property of DP is post-processing immunity: any function applied to the output of a differentially private mechanism cannot weaken its privacy guarantees. This property ensures that subsequent analyses or transformations remain safe without additional noise.

## 2.2. Privacy-preserving Visualizations

Table 2 summarizes representative studies on privacy-preserving visualization. Existing work can be broadly categorized into anonymization and perturbation-based methods, DP mechanisms, utility evaluation studies, specialized visualization types, and system-level frameworks, which are discussed in the following paragraphs.

*Anonymization and perturbation-based approaches.* Early studies, such as Wang et al. (2018) and Chou et al. (2017), adopted an “anonymization” model to guarantee privacy. Such syntactic models, however, can still leak information through composition attacks (Narayanan and Shmatikov, 2008; Dwork and Roth, 2014; Ganta et al., 2008). Avraam et al. (2021) later proposed probabilistic anonymization, which introduces random noise to individual values for creating privacy-preserving visualizations. Although conceptually related to DP, these methods do not provide a quantitative assessment of privacy risk.

*Algorithmic approaches to differentially private visualization.* DP provides formal guarantees even in the presence of side information and has become the gold standard for privacy protection. Zhang et al. (2016) were among the first to systematically examine the challenges of privacy-preserving data visualization. They consider two-dimensional location data and discretize the space into grid cells, where noise is injected using the Laplace mechanism to generate private count estimates. Their work analyzed visual artifacts introduced by noise and proposed methods to convey uncertainty in private visual outputs. Building on this work, Zhang et al. (2020) investigate the application of the Laplace mechanism across multiple visualization formats, including bar charts, and analyze methods for regulating noise to improve visual utility. This approach mitigates distortion while maintaining privacy guarantees. In a related study, Lee (2017) qualitatively examined the effects of Laplacian noise on various visualization types, such as bar graphs, pie charts, and other graphical formats. Using smart grid electricity consumption timeseries data, the study explored how DP constraints affect visual cues and identified pre- and post-processing methods for restoring them.

*Evaluation of visual perception and utility under DP.* Panavas et al. (2023) focused on evaluating the perceptual utility of differentially private scatter-plots. Their comparative study showed that factors such as algorithm choice (e.g., AHP, DAWA), privacy level, bin number, data distribution, and user

task all influence the visual outcome. To address the lack of guidance in balancing these factors, they had experts evaluate 1200 DP scatterplots to assess their effectiveness in preserving aggregate patterns. The results established a ground truth for visual utility and identified multi-scale structural similarity (MS-SSIM) as an effective objective metric for optimizing parameter selection in DP scatterplots. [Liu et al. \(2019\)](#) examined the common belief that aggregating individual data into composite representations such as heatmaps protects privacy. However, their analysis of noise-free heatmaps using eye-tracking data showed that they do not guarantee privacy. To address this, they proposed two noise mechanisms that ensure privacy and analyzed their privacy-utility tradeoff. Their results indicated that the Gaussian noise mechanism, compared to the Laplace mechanism, provides a more effective solution for preserving privacy in heatmaps. These findings have significant implications for the development of differentially private mechanisms in the eye-tracking domain.

*Extensions to specialized visualization types.* More recent work, motivated by [Liu et al. \(2019\)](#) and [Steil et al. \(2019\)](#), has looked at specific visualizations such as heatmaps and boxplots. [Ghazi et al. \(2023\)](#) presented an efficient DP algorithm for heatmaps. The core of their algorithm was a DP procedure that took a set of distributions as input and produced an output close to the average of the inputs in terms of earth mover’s distance. They also proved theoretical bounds on the error of their algorithm under a sparsity assumption, demonstrating that these bounds were near-optimal. [Ramsay and Diaz-Rodriguez \(2024\)](#) introduced a differentially private boxplot and evaluated its effectiveness in displaying location, scale, skewness, and tails of a given empirical distribution. They theoretically demonstrated that the location and scale of the boxplot are estimated with optimal sample complexity, and skewness and tails are estimated consistently. This is not always the case for a boxplot naively constructed from a single existing differentially private quantile algorithm.

*System-level frameworks for DP visualization.* Prior work by [Nanayakkara et al. \(2022\)](#) and [Budiu et al. \(2022\)](#) explores interactive tools that expose the trade-off between privacy budget and accuracy, primarily to facilitate privacy management rather than to support visualization tasks. [Zhou et al. \(2022\)](#) introduced the PriVis model, a Bayesian network-based DP framework, to generate private scatter, line, and bar charts while preserving user-preferred patterns, addressing a limitation of prior work, which often targeted

specific visualization types or algorithms without systematically considering the importance of data patterns in the visual output. They also developed the DPVisCreator system to assist data custodians in implementing this approach and evaluated its effectiveness through quantitative analysis of pattern utility, case studies, and semi-structured expert interviews.

*Limitations of prior work and motivation for our study.* Despite these advances, prior work mainly focused on generic visualization types rather than domain-specific cases such as brain functional connectivity, which may require different techniques. Most studies do not systematically compare DP mechanisms to minimize required noise, and pre- and/or post-processing methods often lack structured workflows, limiting their applicability to neuroimaging. Evaluations also rely heavily on qualitative assessment without exploring potential quantitative approaches to measure visual utility. In contrast, our work focuses on functional connectivity in neuroimaging and provides a systematic evaluation of DP mechanisms, composition, and pre- and/or post-processing methods.

### 2.3. Functional Connectivity in Neuroimaging

In this work, we focus on neuroimaging data, particularly functional magnetic resonance imaging (fMRI), due to its sensitivity, high dimensionality, and collaborative use across institutions. fMRI data can reveal aspects of an individual’s cognitive state, making privacy protection critical. DP offers a principled framework for secure data sharing in such settings. Furthermore, the variety of visualization modalities employed in fMRI studies enables a comprehensive examination of privacy-preserving visualization techniques, with insights that can extend to other neuroimaging modalities and scientific domains.

Resting-state fMRI characterizes spontaneous blood oxygenation level dependent (BOLD) signal fluctuations when subjects are scanned in the absence of externally imposed tasks. A series of MRI images is acquired over time, with each image recording BOLD signal intensity at each voxel. We denote the BOLD value for voxel  $v$  at time  $t$  as  $S(v, t)$ . Typical MRI images contain roughly 100,000 voxels, which can be grouped into regions of interest (ROIs) using various parcellation schemes (Tzourio-Mazoyer et al., 2002; Craddock et al., 2012; Gordon et al., 2016; Du et al., 2020). ROI timeseries are obtained by averaging voxel-level signals within each ROI:  $S(r, t) = \frac{1}{L} \sum_{i=1}^L S(v_i, t)$ , where  $r$  is the ROI and  $L$  is the number of voxels in that ROI. Analyses

| Study                            | Data Type                                                   | Visualization                                                            | Privacy Method                                                                                       | Advantages                                                                                                                         | Limitations                                                                                                                          |
|----------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Zhang et al. (2016)              | 2D spatial location data                                    | Spatial heatmaps                                                         | DP via grid-based injection of calibrated Laplace noise                                              | First systematic study of DP visualization artifacts.                                                                              | Injected noise introduces structural distortions, reducing visualization utility and readability.                                    |
| Lee (2017)                       | Smart-grid timeseries data                                  | Bar charts, pie charts, scatterplots, and heatmaps                       | Laplace mechanism                                                                                    | Evaluates the visual impact of Laplace noise and characterizes the utility-privacy tradeoff across multiple chart types.           | Primarily qualitative and limited to low-dimensional statistical data.                                                               |
| Chou et al. (2017)               | Ontological networks with multi-attribute categorical nodes | Color-coded node-link diagrams with edge bundling and adjacency matrices | Relational $k$ -anonymity and $l$ -diversity via node merging, edge bundling, and perceptual masking | Mitigates identity disclosure in multi-attribute graphs and obscures sensitive categorical links without adding random noise.      | Vulnerable to background knowledge attacks; perceptual masking does not prevent algorithmic information leakage.                     |
| Wang et al. (2018)               | Unweighted and undirected graph topology                    | Node-link diagrams and degree-distribution visualizations                | Topological $k$ -anonymity against degree, hub, and sub-graph attacks via edge addition              | Preserves structural properties deterministically via edge addition while maintaining global network density.                      | Lacks strong security bounds and is vulnerable to linkage attacks when attackers have auxiliary structural knowledge of the network. |
| Liu et al. (2019)                | Eye-tracking data                                           | Heatmaps                                                                 | Gaussian and Laplace mechanisms                                                                      | Shows that aggregated heatmaps do not inherently preserve privacy and maintains visual smoothness through DP aggregation.          | Limited to independent 2D spatial coordinates.                                                                                       |
| Zhang et al. (2020)              | Tabular data                                                | Bar charts, pie charts, line charts, and scatterplots                    | DP via Laplace and Gaussian noise mechanisms                                                         | Empirically shows that summary-level visual tasks maintain high accuracy under DP noise and identifies robust visualization types. | Shows severe distortion in local value-retrieval tasks and fails to support expert analysis of high-dimensional data.                |
| Avraam et al. (2021)             | Continuous numerical microdata                              | Histograms, scatterplots, heatmaps, boxplots, and regression plots       | $k$ -anonymization, deterministic $k$ -NN clustering, and probabilistic perturbation                 | Retains statistical distributions, regression trends, and visualization utility with minimal heuristic parameters.                 | Fails to provide formal privacy guarantees, making it susceptible to reconstruction and membership inference attacks.                |
| Zhou et al. (2022)               | Tabular data                                                | Multi-chart dashboards with scatterplots, line and bar charts            | Bayesian network-based DP                                                                            | Introduces user-preferred constraints to improve structural fidelity of DP synthetic data across multiple charts.                  | Sensitive to data size for structure control and lacks real-time feedback for high-dimensional data.                                 |
| Panavas et al. (2023)            | 2D continuous data                                          | Scatterplots                                                             | AHP, DAWA, and DP variants                                                                           | Provides empirical guidance for selecting DP algorithms and binning parameters to improve visual utility under expert evaluation.  | Limited to traditional 2D continuous plots and lacks scalability for high-dimensional matrix analysis.                               |
| Ghazi et al. (2023)              | 2D geographic continuous data                               | Heatmaps                                                                 | Pyramidal transform with adaptive Laplace noise                                                      | Achieves near-optimal $\epsilon$ -DP under EMD while preserving visual utility at high grid resolutions.                           | Relies on data sparsity assumptions and is not directly applicable to the local DP model.                                            |
| Ramsay and Diaz-Rodriguez (2024) | 1D continuous data                                          | Boxplots                                                                 | DP quantile estimation                                                                               | Achieves optimal sample complexity for inner box quantiles and consistent extreme quantiles for whiskers under centralized DP.     | Fails to capture individual outliers and lacks optimal budget allocation for sequential iterative exploratory workflows.             |

Table 2: Summary of prior work on privacy-preserving visualization.



at the voxel and ROI levels provide complementary insights into brain functional connectivity, motivating diverse visualization approaches to effectively present these results.

FNC quantifies inter-regional connectivity based on correlations in neural activity signals (Mohanty et al., 2020). Various methods exist to compute these correlations (Marrelec et al., 2006; Hlinka et al., 2011); in this work, we typically use the Pearson correlation coefficient, which ranges from  $-1$  to  $1$  and measures the linear relationship between ROI-based timeseries. For any two ROIs, the correlation is given by

$$X_{ij} = \frac{\sum_{t=1}^T S(r_i, t)S(r_j, t)}{\sqrt{\sum_{t=1}^T S^2(r_i, t)}\sqrt{\sum_{t=1}^T S^2(r_j, t)}},$$

where  $T$  denotes the number of timepoints and  $S(\cdot, t)$  is assumed to be centered at 0. The FNC for a subject  $n$  across  $r$  brain regions is represented by an  $r \times r$  adjacency matrix  $X^{(n)}$ , with each entry corresponding to the correlation between a pair of regions. We focus on visualizing the average FNC across a group of  $N$  subjects (see Section 3.1), defined as

$$\bar{X} = \frac{1}{N} \sum_{n=1}^N X^{(n)}.$$

For visualization, each entry of  $\bar{X}$  is first scaled to  $[0, 1]$ , then discretized into a finite color set  $\mathcal{C}$ , and finally represented as an RGB value  $(R_{ij}, G_{ij}, B_{ij})$ . In addition to visualizing the average FNC matrix, a connectogram can be used to represent FNC results graphically, with nodes as brain regions and edges indicating connection strength (see Section 3.2).

While FNC captures large-scale network interactions using ROI-based timeseries, SBC provides detailed information about the connectivity of a specific seed region with the rest of the brain. SBC involves selecting a seed voxel or ROI and computing its correlation with the timeseries of all other brain voxels. The resulting connectivity patterns are typically visualized as SBC spatial maps (Joel et al., 2011) (see Section 4).

#### 2.4. Challenges in DP Functional Connectivity Visualization

When applying DP to visualization, several key challenges arise, including the mismatch between visualization-specific computations and generic DP



methods, repeated data access in exploratory analysis workflows, and the difficulty of defining utility for visual outputs (Bhattacharjee et al., 2020; Zhang et al., 2016; Zhou et al., 2022). When thinking about visualization, the notion of “utility” can differ from standard quantitative metrics. In some scenarios, a private visualization has high utility if it enables an expert to reach the same qualitative conclusions as those obtained from the non-private visualization.

To address these challenges, we take the following approaches. To resolve the mismatch between visualization requirements and generic DP approaches, we construct FNC-specific mechanisms that produce private correlation matrices for downstream visualization. For connectograms, we design a differentially private workflow for group comparisons, and for SBC visualization, we propose a DP procedure to systematically identify and present statistically significant voxel-wise patterns. To manage multiple data accesses, we employ (adaptive) composition techniques (Kairouz et al., 2015; Dong et al., 2020, 2022; Mironov, 2017) to quantify and control cumulative privacy risk. Finally, to address the challenge of utility, we provide both qualitative assessments and quantitative measurements.

In addressing these issues, we outline a systematic process for designing privacy-preserving visualizations across different visualization types and application domains, emphasizing both visualization quality and data privacy. Although our focus is on neuroimaging, the *process* we outline can be extended to other domains. Visualization typically involves computing or aggregating statistics from raw data through multiple steps. The first step is to identify the statistics or aggregated quantities needed for visualization. Next, suitable pre-processing strategies are applied to reduce the noise required for privacy protection. Different visualization types may benefit from different pre-processing techniques. We then design the differentially private mechanism, which may involve selecting appropriate noise distributions, applying composition methods, or constructing privacy-preserving workflows. Finally, post-processing methods are used to mitigate the effects of perturbation. Following this process, we demonstrate how privacy-preserving visualizations are constructed in neuroimaging (see Sections 3 and 4).

### 3. FNC Visualization

#### 3.1. FNC Matrix Visualization

In this section, we describe the privacy-preserving visualization process for the average FNC matrix, summarized in Table 3, together with pre- and post-processing techniques that enhance visual utility.

| Algorithm    | Name                | Strategy         | Mechanism   | Composition |
|--------------|---------------------|------------------|-------------|-------------|
| matrix-level | Pure                | add perturbation | Laplace     | —           |
|              | Approx              | add perturbation | Gaussian    | —           |
|              | Exp                 | random mapping   | Exponential | —           |
| entry-level  | Pure <sub>k</sub>   | add perturbation | Laplace     | exact       |
|              | Pure <sub>r</sub>   | add perturbation | Laplace     | RDP-based   |
|              | Approx <sub>k</sub> | add perturbation | Gaussian    | exact       |
|              | Approx <sub>r</sub> | add perturbation | Gaussian    | RDP-based   |
|              | Exp <sub>k</sub>    | random mapping   | Exponential | exact       |

Table 3: A summary of the mechanisms explored in this paper is provided. Although other noise distributions may perform better for specific tasks, we focus on Laplace and Gaussian mechanisms, which are the most widely studied, for clarity of exposition.

##### 3.1.1. Matrix-level Differentially Private Algorithms

In FNC matrix visualization, the outputs correspond to RGB values associated with each pairwise correlation in the averaged matrix. To guarantee DP, we adopt the following strategy: first compute the average FNC matrix under DP, then apply standard quantization and color mapping. This approach can be viewed as an instance of *output perturbation*, where noise is applied directly to the averaged FNC matrix:

$$\tilde{X} = \bar{X} + Z,$$

where the choice of noise  $Z$  is determined by the privacy parameters. The perturbed values  $\tilde{X}$  are clipped to lie within the interval  $[-1, 1]$ . We consider the following two mechanisms:

- **Pure:**  $(\epsilon, 0)$ -DP is achieved using the Laplace mechanism, where noise  $Z$  is drawn from  $\text{Laplace}(0, \Delta_1/\epsilon \cdot I)$ . Here,  $\Delta_1 = r(r-1)/N$  denotes the  $\ell_1$  sensitivity of  $\bar{X}$ .

- **Approx:**  $(\epsilon, \delta)$ -DP is achieved using the analytic Gaussian mechanism, where noise  $Z$  is drawn from  $\mathcal{N}(0, \sigma^2 \cdot I)$ . The standard deviation  $\sigma$  is chosen as the minimum value satisfying the condition in (Balle and Wang, 2018, Theorem 8), based on the  $\ell_2$  sensitivity  $\Delta_2 = \sqrt{2r(r-1)}/N$  of  $X$ .

### 3.1.2. Entry-level Differentially Private Algorithms

For matrix-level approaches, we first determine the sensitivity of the FNC matrix and then construct the corresponding perturbed version. To improve efficiency, we apply composition at the entry level. Let  $(\epsilon, \delta)$  denote the overall privacy budget, and let  $k = r(r-1)/2$  be the number of entries. We then examine the following variants:

- **Pure<sub>k</sub>:** We allocate per-entry privacy parameters  $(\epsilon', 0)$  via exact composition and perturb each entry independently using the Laplace mechanism with sensitivity  $\Delta = 2/N$ . The privatized entries are given by  $\tilde{X}_{ij} = \bar{X}_{ij} + Z_{ij}$ , where  $Z_{ij} \sim \text{Laplace}(0, \Delta/\epsilon')$ . For one-dimensional outputs, the global  $\ell_1$  and  $\ell_2$  sensitivities are identical, so we use  $\Delta$  to denote the sensitivity without distinction between norms.
- **Pure<sub>r</sub>:** Given the overall privacy budget, we determine the Laplace scale parameter  $b$  using RDP-based composition. Each entry is then perturbed independently as  $\tilde{X}_{ij} = \bar{X}_{ij} + Z_{ij}$ , where  $Z_{ij} \sim \text{Laplace}(0, b)$ .
- **Approx<sub>k</sub>:** We allocate per-entry privacy parameters  $(\epsilon', \delta')$  via exact composition and perturb each entry independently using the analytic Gaussian mechanism with sensitivity  $\Delta = 2/N$ . The privatized entries are given by  $\tilde{X}_{ij} = \bar{X}_{ij} + Z_{ij}$ , where the variance of  $Z_{ij}$  is chosen to satisfy the condition in (Balle and Wang, 2018, Theorem 8).
- **Approx<sub>r</sub>:** We determine the variance  $\sigma^2$  of the Gaussian noise using RDP-based composition under the given privacy parameters. Each entry is then perturbed independently as  $\tilde{X}_{ij} = \bar{X}_{ij} + Z_{ij}$ , where  $Z_{ij} \sim \mathcal{N}(0, \sigma^2)$ .

In many visualization workflows, especially in exploratory analysis, the same dataset may be accessed repeatedly to compute different intermediate statistics, each of which contributes to the final visual output. This repeated access necessitates careful accounting of privacy loss, making composition an essential consideration. Under “pure” differential privacy ( $\delta = 0$ ), the

matrix-level Laplace mechanism provides strong guarantees, but often requires injecting a substantial amount of noise. By relaxing the guarantee to “approximate” differential privacy ( $\delta > 0$ ), composition techniques can be leveraged to distribute the privacy budget more effectively, thereby reducing the magnitude of perturbation. In addition, we examine the performance differences between matrix-level and entry-level analytic Gaussian mechanisms under this setting.

We also explored the Exponential mechanism (McSherry and Talwar, 2007), which performs a randomized mapping from  $X$  to the set of all RGB images  $\mathcal{H}$ , either as a whole (denoted  $\text{Exp}$ ) or entry-by-entry (denoted  $\text{Exp}_k$ ). The  $\text{Exp}$  method provides  $(\epsilon, 0)$ -DP by sampling  $H \in \mathcal{H}$  according to the utility function  $u(\bar{X}, H) = -\|\bar{X} - H\|_2$ , but this is computationally infeasible due to the size of  $\mathcal{H}$ . The entry-level approach  $\text{Exp}_k$  applies an  $(\epsilon', 0)$  mechanism with exact composition to assign each  $\bar{X}_{ij}$  a color  $C \in \mathcal{C}$  based on a utility score defined as  $-|\bar{X}_{ij} - C|$ , with sensitivity  $2/N$ . However, this approach yielded substantially poorer visual quality compared to output perturbation-based methods.

### 3.1.3. Pre-processing Approaches

To reduce the amount of noise required under a given privacy budget  $(\epsilon, \delta)$ , we consider a simple pre-processing step that restricts the range of the data. In FNC, extremely large correlation values (e.g.,  $|\bar{X}_{ij}| \geq 0.9, i \neq j$ ) are rarely observed in practice. This observation motivates clipping  $\bar{X}_{ij}$  to a bounded interval  $[-b, b]$  with  $0 < b < 1$ . Under this constraint, the sensitivities of the averaged FNC matrix scale with  $b$ , yielding  $\Delta_1 = br(r - 1)/N$  and  $\Delta_2 = b\sqrt{2r(r - 1)}/N$ , while the sensitivity of each entry becomes  $\Delta = 2b/N$ .

In this work, we adopt a clipping method (Near and Abuah, 2021) to estimate suitable bounds for the FNC values (see Algorithm 1). The procedure evaluates a sequence of candidate bounds by progressively increasing  $b$  within  $[0, 1]$ . For each candidate bound, values of  $X_{ij}^{(n)}$  that fall outside  $[-b, b]$  are truncated to the nearest boundary. For each choice of  $b$ , we compute the aggregate sum of the clipped entries and perturb it using the Laplace mechanism to ensure privacy. The final bound is selected based on the stability of the resulting privatized sums (see Figure 3).

Bounds may also be estimated through several alternative methods, including domain-specific knowledge, publicly available data, or other privacy-preserving estimation procedures. For example, public datasets can help

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**Algorithm 1** Method for choosing the clipping bound  $b$ 


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**Input:** A group of FNC matrices  $X^{(1)}, X^{(2)}, \dots, X^{(N)}$ , privacy budget  $\epsilon_b$  for each possible bound

**Output:** A list of  $\{\tilde{s}_b\}$  for all possible  $b$

**for**  $b = 0.1, 0.2, \dots, 1$  **do**

1.  $s_b = 0$

2. **for**  $i = 1, \dots, r; j > i; n = 1, \dots, N$  **do**

    If  $|X_{ij}^{(n)}| \leq b$ :  $s_b = s_b + X_{ij}^{(n)}$ ;

    If  $X_{ij}^{(n)} > b$  or  $X_{ij}^{(n)} < -b$ :  $s_b = s_b + b$  or  $s_b = s_b - b$ , respectively.

3.  $\tilde{s}_b = s_b + \text{Laplace}(0, br(r-1)/\epsilon_b)$

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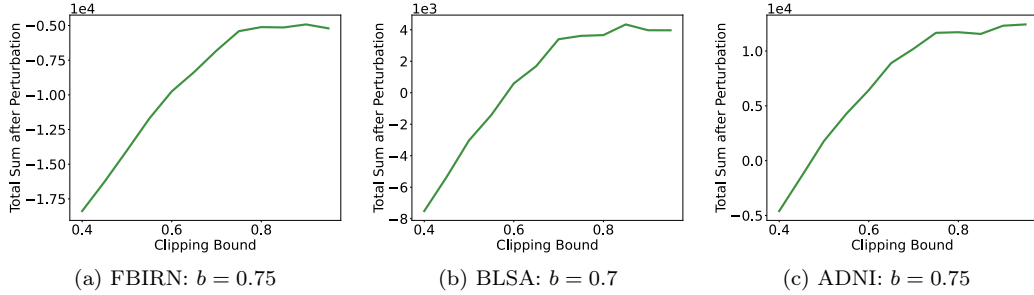


Figure 3: Clipping method for choosing  $b$  on three datasets. The total privacy budget allocated for bound estimation is 0.5. The search for  $b$  begins at 0.4 rather than 0, assuming that the bound of FNC values exceeds 0.4.

determine upper and lower bounds. Another approach is to estimate differentially private quantiles and use them as bounds (McSherry and Talwar, 2007; Smith, 2011). Liu et al. (2019) propose an algorithm that optimizes the expected mean squared error between perturbed aggregated data and its original counterpart to estimate bounds. All approaches involve a trade-off between information loss and the level of noise required to ensure differential privacy. Tighter bounds reduce sensitivity and therefore require less noise to ensure DP. However, overly restrictive clipping may discard a substantial portion of the underlying information.

### 3.1.4. Post-processing Approaches

A key property of differential privacy is its invariance to post-processing, which ensures that applying data-independent transformations to private

outputs does not weaken the privacy guarantees. This property allows additional processing steps to be applied after noise injection without consuming further privacy budget. In practice, such post-processing can be used to refine visual outputs, helping to reduce artifacts introduced by noise in FNC visualizations and improve their overall interpretability.

To improve the visual quality of privatized FNC matrices, we apply two post-processing strategies based on Singular Value Decomposition (SVD) and Haar wavelet transform (Mulcahy, 1997). These methods are motivated by the observation that FNC matrices contain structured connectivity patterns, whereas injected DP noise is largely unstructured. As a result, meaningful signal can often be represented using a compact set of dominant components, enabling denoising while preserving the primary network organization.

For the SVD approach, we decompose the noisy matrix  $\tilde{X}$  into  $U\Sigma V^\top$  and retain only the dominant singular components. The number of retained components is determined by examining the decay pattern of singular values. The resulting low-rank approximation  $\tilde{X}_s = U_s \Sigma_s V_s^\top$  is then used for visualization. For the Haar wavelet transform method, we first transform  $\tilde{X}$  into the wavelet domain via  $W = P^\top \tilde{X} P$ , where  $P$  is the Haar transformation matrix. In this domain, many coefficients tend to be small in magnitude, allowing for effective noise suppression through thresholding. After applying coefficient shrinkage to obtain  $W_w$ , we reconstruct the processed matrix using the inverse transform, yielding  $\tilde{X}_w = (P^\top)^{-1} W_w P^{-1}$  for visualization.

### 3.1.5. Evaluation Metrics

Quantitatively assessing privacy-preserving visualizations is challenging due to the lack of a universal definition of visual utility (Zhang et al., 2016). In this work, we explore a set of quantitative metrics. We adopt three complementary metrics that capture pointwise accuracy, structural organization, and local smoothness.

We first measure pointwise numerical deviation between the original average FNC matrix and its privatized counterpart using MSE. MSE provides a direct assessment of element-wise accuracy and reflects overall perturbation magnitude. However, it does not capture higher-level structural properties of the connectivity matrix.

Since FNC matrices are interpreted through block-structured functional networks, we additionally evaluate network modularity (Rubinov and Sporns, 2011). Modularity measures the strength of community structure relative to a random null model. A reduction in modularity after privacy perturba-

tion indicates that intrinsic functional modules become less distinguishable, reflecting structural degradation and reduced preservation of visually meaningful connectivity patterns.

To quantify local fluctuation patterns introduced by privacy noise, we compute the total variation (TV) (Michel et al., 2011) of the connectivity matrix and report the TV ratio between the private and original matrices. TV measures the aggregate magnitude of local differences between adjacent matrix entries, thereby characterizing high-frequency perturbations. A higher TV ratio indicates increased local irregularity and reduced smoothness of the connectivity pattern, suggesting greater visual distortion introduced by privacy perturbation.

To further assess perceptual utility, we additionally conduct an expert-based qualitative evaluation, as described in Section 5.

### 3.2. Connectogram Visualization

#### 3.2.1. Non-private Workflow for Group-Level Comparison

Connectograms are commonly used to visualize group-level differences in brain connectivity patterns, for example when comparing healthy controls (HC) and schizophrenia (SZ) patients. The construction of a connectogram typically involves transforming the FNC values using Fisher’s  $Z$ -transformation, followed by statistical testing to identify significant group differences. In particular, we perform a two-tailed two-sample  $t$ -test with Bonferroni correction ( $p < 0.05$ ) to assess the significance of connectivity differences between the two groups (Du et al., 2020).

#### 3.2.2. Private Workflow for Group-Level Comparison

We develop a differentially private workflow for connectogram visualization tailored to group comparison tasks. The framework consists of two components. The first component protects FNC values, while the second component protects the statistical quantities used to identify significant group differences. Prior studies have shown that releasing aggregate statistics and inference results may enable re-identification or membership inference of individuals (Homer et al., 2008; Gaboardi et al., 2016; Vu and Slavkovic, 2009). This suggests that protecting FNC values alone is not sufficient if statistical measures derived from the same data remain publicly accessible. Consequently, a privacy-preserving workflow for connectogram construction must also account for the statistical procedures used to identify group differences.

Rather than directly perturbing statistical quantities such as  $p$ -values or  $t$ -statistics, we focus on designing a more stable alternative under privacy constraints. Direct perturbation of these statistics is problematic, as their sensitivities are challenging to quantify and small perturbations can lead to large changes in  $p$ -values. To address these challenges, we instead employ a nonparametric testing framework, which is empirically more robust under DP. In particular, we adopt a private workflow based on the absolute Kruskal-Wallis (AKW) test (Couch et al., 2019) for constructing connectograms.

Consider a database of size  $N$  partitioned into  $g$  groups. Let  $N_i$  denote the size of group  $i$ , and  $r_{ij}$  the rank of the  $j$ -th element in group  $i$ . Define the mean rank of group  $i$  as  $\bar{r}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} r_{ij}$ , and the overall mean rank as  $\bar{r} = \frac{N+1}{2}$ . The absolute Kruskal-Wallis  $h$ -statistic is given by

$$h = (N - 1) \frac{\sum_{i=1}^g N_i |\bar{r}_i - \bar{r}|}{\sum_{i=1}^g \sum_{j=1}^{N_i} |r_{ij} - \bar{r}|}$$

and has a sensitivity of 8 (Couch et al., 2019, Algorithm  $\widetilde{\text{KW}}_{\text{P}}$ ).

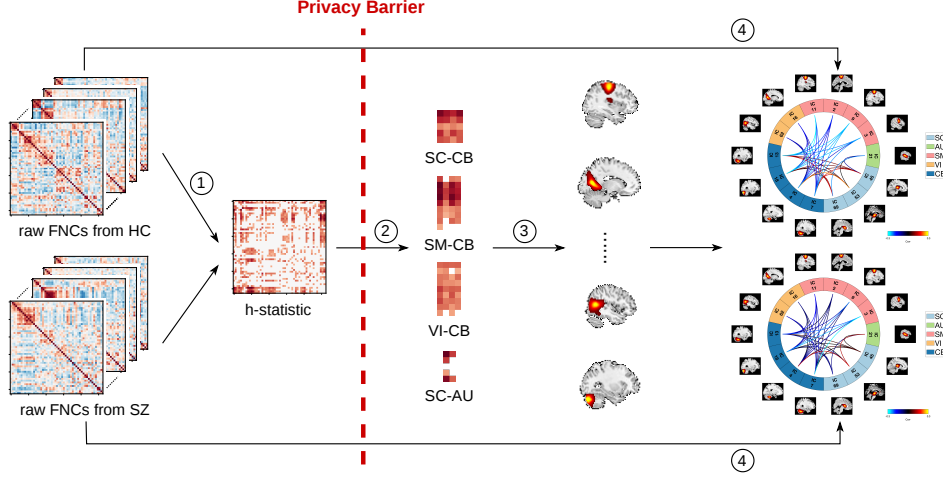


Figure 4: Privacy-preserving workflow for connectogram visualization. The privacy barrier refers to the stage where sensitive data (e.g., FNC values,  $h$ -statistics) are perturbed using differential privacy techniques. Data after the privacy barrier is considered privacy-preserving, and it is impossible to infer the true data from this released information.



Our workflow (see Figure 4) proceeds as follows: (1) Compute the AKW test on FNC matrices from two groups to obtain an  $h$ -statistic matrix. (2) For each  $h$ -statistic submatrix, which is partitioned by brain domains (Du et al., 2020, Table 2), calculate the average  $h$ -statistic. Apply the Report Noisy Max (RNM) algorithm (Dwork, 2006) to select the top  $p_d$  submatrices in a differentially private manner. The RNM algorithm is a DP technique for private selection. Instead of selecting the top-ranked items based on their true values, RNM adds calibrated noise to each value and selects the top-ranked items based on the noisy values. The number of selected items is determined by the privacy budget allocated at this stage. (3) For each selected brain domain, we aggregate the  $h$ -statistics at the region level and apply the RNM mechanism to privately identify the top  $p_r$  regions. (4) We then derive privatized connectivity values from the group-level average FNC matrices. These privacy-preserving nodes and edges are subsequently used to construct the connectogram for group comparison.

## 4. SBC Visualization

### 4.1. Non-private Workflow for Significant Connectivity

SBC characterizes the connectivity pattern between a predefined seed and the entire brain. This involves computing Fisher-transformed bivariate correlation coefficients between the seed BOLD timeseries and the timeseries of each individual voxel (Joel et al., 2011). The correlation between the seed and any voxel  $v$  is defined as

$$\frac{\sum_{t=1}^T S(\cdot, t)S(v, t)}{\sqrt{\sum_{t=1}^T S^2(\cdot, t)}\sqrt{\sum_{t=1}^T S^2(v, t)}},$$

where  $S(\cdot, t)$  represents the BOLD timeseries from the seed (either a single voxel or an ROI) at time  $t$ , and  $T$  is the number of timepoints in the experiment. The seed timeseries  $S(\cdot, t)$  is assumed to be mean-centered. The correlation coefficient is then converted to a  $z$ -score using the Fisher  $Z$ -transform. A one-sample  $t$ -test is subsequently performed to determine whether the mean connectivity across subjects significantly differs from zero, indicating significant positive or negative connectivity.

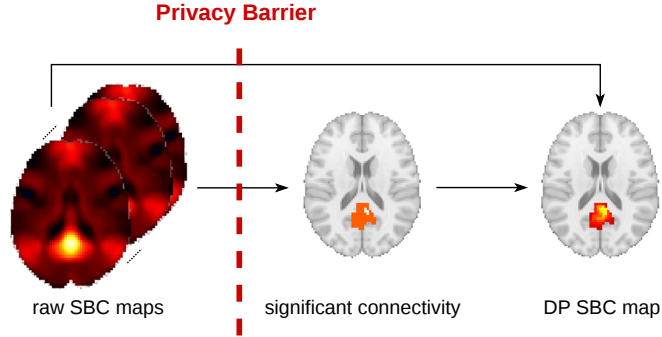


Figure 5: Privacy-preserving workflow for SBC map visualization.

#### 4.2. Private Workflow for Significant Connectivity

The privacy-preserving SBC visualization process is quite similar to connectogram visualization and also involves two main components (see Figure 5): selecting significant connectivities in a privacy-preserving manner and displaying  $z$ -scores with differential privacy. To identify statistically significant connectivities, we employ the Wilcoxon signed-rank (WSR) test, a nonparametric alternative to the one-sample  $t$ -test, as nonparametric tests are often more effective than their parametric counterparts. For a given database, let  $s_i$  denote the sign of the data  $x_i$ , and let  $r_i$  represent the rank of  $|x_i|$ , determined by the magnitude of the data. The WSR test statistic is then calculated as  $w = \sum s_i r_i$ . Connectivities are selected at a significance level of  $p < 0.05$  after applying Bonferroni correction (Couch et al., 2019, Algorithm  $\widetilde{\text{WP}}_P$ ).

To protect the  $z$ -scores, we use the analytic Gaussian mechanism to generate noise, which is added to the correlations before applying the Fisher  $Z$ -transform. Although narrowing the range of correlations could reduce the required noise, this approach is not suitable here because high correlations are expected (see Figure 6a). The privacy-protected correlation map exhibits noticeable salt-and-pepper noise (see Figure 6b), which is absent in FNC visualization. One likely reason is that connections within the same brain network tend to be strong, resulting in relatively stable and similar correlation values. To mitigate this noise, we apply a median filter (Erkan et al., 2018) as a post-processing method (see Figure 6c). After identifying significant connectivities and their corresponding values, we construct the

privacy-preserving SBC map.

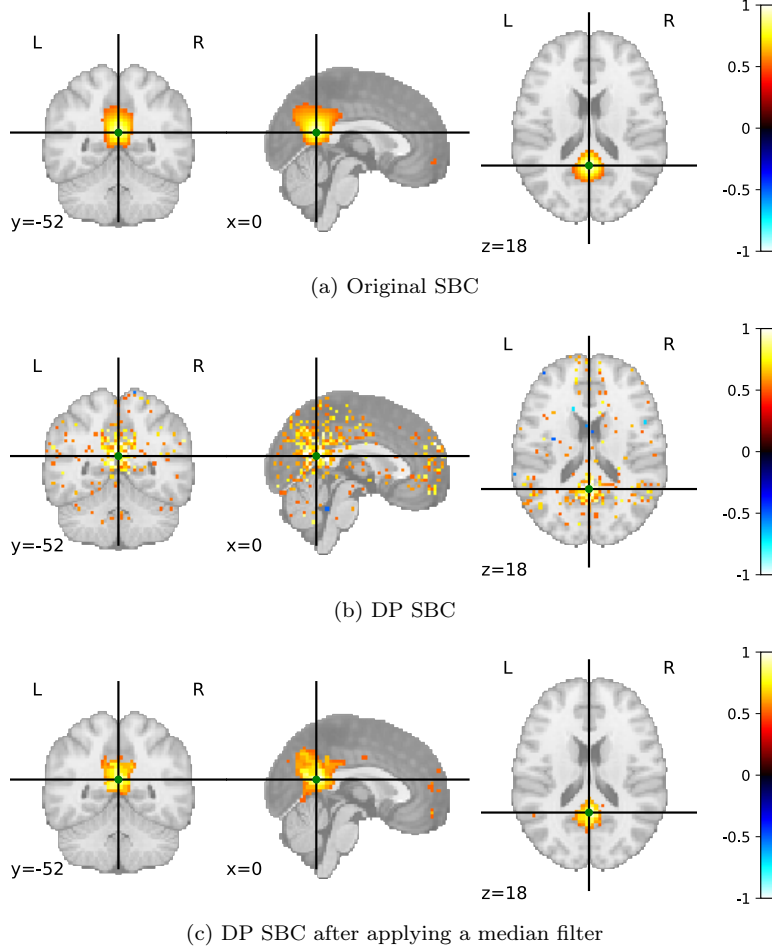


Figure 6: SBC maps on the NITRC dataset using the Posterior Cingulate Cortex (PCC) as the seed (indicated by the green dot), with parameters  $N = 1082$ ,  $\epsilon = 2.5$ , and  $\delta = N^{-3/2}$ . Correlations with absolute values greater than 0.5 are displayed.

## 5. Experimental Results

We aim to investigate the following research questions: (1) Which algorithm most effectively preserves functional connectivity patterns under the same level of privacy? (2) Can pre- and post-processing methods enhance visualization quality? (3) To what extent does the differentially private connectogram workflow recover the same significant brain regions and connections

as the non-private workflow? (4) Is the differentially private workflow for SBC visualization as effective as its non-private counterpart in identifying significant connectivities? (5) What is the relationship between numerical utility metrics and perceived visualization quality?

We conducted privacy-preserving FNC visualization experiments using datasets from the Function Biomedical Informatics Research Network (FBIRN), the Baltimore Longitudinal Study of Aging (BLSA), and the Alzheimer’s Disease Neuroimaging Initiative (ADNI) (Du et al., 2020; Sendi et al., 2021; Ferrucci, 2008). Our findings were consistent across all datasets. For connectogram visualizations, we used the FBIRN dataset to compare HC and SZ groups. For SBC visualizations, we employed the NITRC dataset.

### 5.1. Methods Comparison

In general, we found that Gaussian mechanisms preserved visual utility better than Laplace mechanisms, which in turn outperformed Exponential mechanisms (see Figure 7, Figure A.3 and Figure A.4). Overall, Approx-Clip-SVD achieved the best performance.

This superior performance can be explained from both structural and theoretical perspectives. At a structural level, the privacy-preserving FNCs produced by this method retain the same block-structured organization seen in the original matrix, with recognizable within-network modules along the diagonal and comparatively weaker cross-network structure off-diagonal. The method preserves the relative ordering of strong versus weak connectivity patterns across domains, so the same major network-network relationships remain visually interpretable even though individual entries show slight deviations. In contrast, less effective DP examples introduce noise that partially blurs or fragments these modules, reducing contrast and making the within-network blocks less coherent, which can obscure higher-level structure.

From a theoretical perspective, the analytic Gaussian mechanism typically introduces less perturbation than the Laplace mechanism in high-dimensional settings, as its noise scale depends on the  $\ell_2$  sensitivity, which is generally smaller than the corresponding  $\ell_1$  sensitivity. This reduction in noise helps preserve structural patterns in the resulting visualizations. In contrast, the Exponential mechanism tends to produce less stable visual outputs due to its high degree of randomness. Under  $\text{Pure}_k$ , the distribution of each perturbed entry is proportional to  $\exp(-N\epsilon'|z_{ij} - \bar{X}_{ij}|/2)$ , whereas under  $\text{Exp}_k$ , the probability of selecting a candidate value  $c$  is proportional to  $\exp(-N\epsilon'|c - \bar{X}_{ij}|/4)$ . The Exponential mechanism is particularly effective

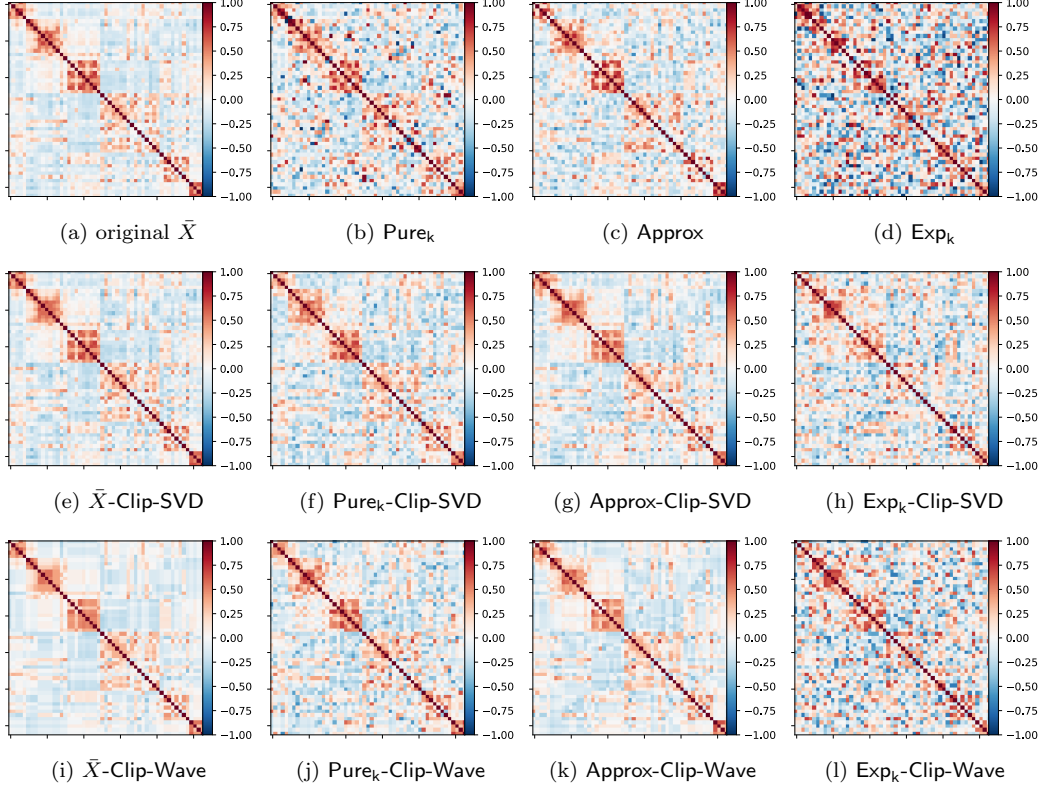


Figure 7: Visualization of the FBIRN dataset with  $N = 1056$ , clipping bound  $b = 0.75$ , number of components used in SVD set to 10, threshold used in the wavelet transform set to 0.2, and number of colors set to 256. The first row represents methods without pre- or post-processing, the second row represents methods with clipping and SVD post-processing, and the third row represents methods with clipping and wavelet post-processing. The overall privacy guarantee for all perturbed FNC matrices is  $(\epsilon = 2, \delta = N^{-3/2})$ -DP. As expected, the Laplace mechanism performs worse than the analytic Gaussian mechanism since the  $\ell_1$  sensitivity is larger than the  $\ell_2$  sensitivity in high dimensions. Even with composition methods applied to reduce noise,  $\text{Pure}_k$  still does not outperform  $\text{Approx}$ . The Exponential mechanism is barely usable. Clipping and SVD post-processing significantly enhance the visual fidelity of private FNC matrices, whereas wavelet post-processing introduces additional visual distortion. This suggests that classical image processing techniques developed for natural images (e.g., photos) may not be suitable for this type of visualization.

in settings where utility is highly sensitive to small changes in the output. However, in FNC visualization, small variations in correlation values often lead to only minor differences in color encoding, making this mechanism less

suitable.

From a composition perspective, both exact and RDP-based approaches provide better noise-privacy trade-offs than advanced composition across different mechanisms. For the Laplace mechanism, distributing the privacy budget at the entry level ( $\text{Pure}_k$ ) leads to lower noise compared to the alternative ( $\text{Pure}_r$ ). In contrast, for the analytic Gaussian mechanism, the entry-level allocation ( $\text{Approx}_k$ ) results in higher noise.

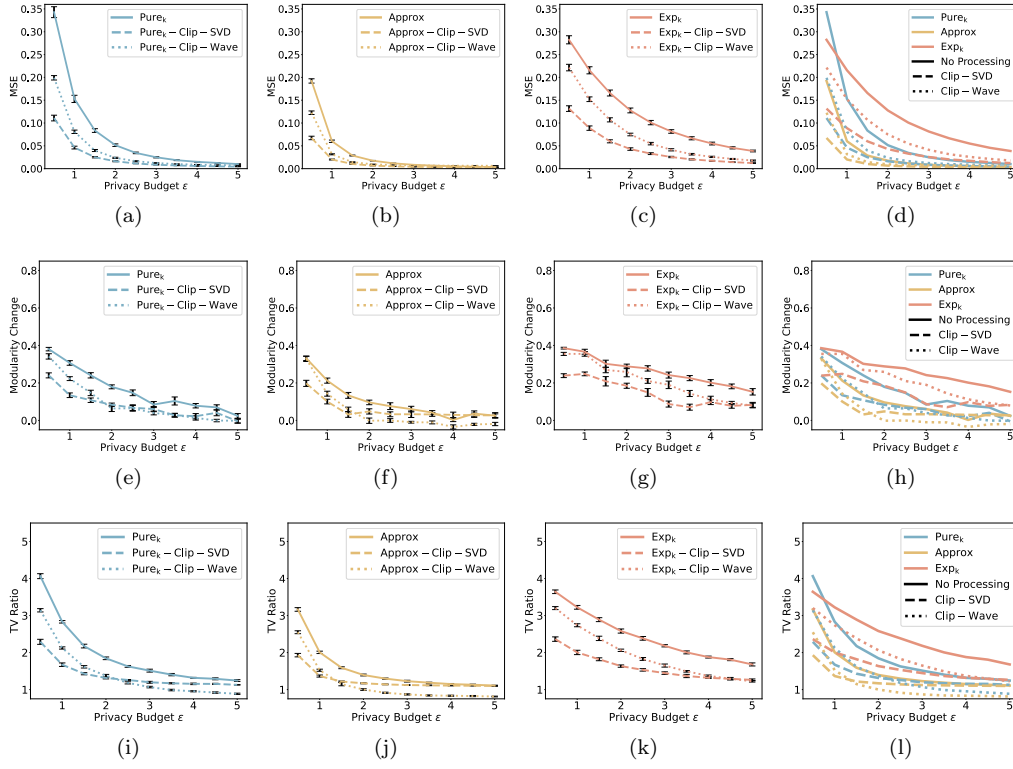


Figure 8: Quantitative evaluation of privacy-preserving FNC visualization on the FBIRN dataset, where the FNC matrices are constructed from 53 intrinsic connectivity networks derived using the NeuroMark pipeline (Du et al., 2020). The first row reports MSE between the original and private average FNC matrices. The second row shows modularity change, defined as  $Q_{\text{original}} - Q_{\text{DP}}$ , where  $Q$  denotes signed Louvain modularity. The third row presents the TV ratio, computed as  $\text{TV}_{\text{DP}}/\text{TV}_{\text{original}}$ . Columns correspond to (a)(e)(i)  $\text{Pure}_k$ -, (b)(f)(j)  $\text{Approx}_k$ -, (c)(g)(k)  $\text{Exp}_k$ -related methods, and (d)(h)(l) all methods. Each configuration is repeated 50 times, and error bars are provided.

### 5.2. Processing Effect

Pre- and post-processing steps play a crucial role in improving the visual quality of privacy-preserving visualizations. In the case of FNC visualization, Figures 7g and 7k correspond to different processing strategies, both of which yield visually faithful representations that closely resemble the original average FNC shown in Figure 7a.

To quantify this effect, we compute the MSE between the original average FNC and its private counterpart. As illustrated in Figure 8 (a)-(d), Figure A.1, and Figure A.2, the impact of pre- and post-processing is most evident when the privacy budget is small, while its effect diminishes as the privacy budget increases. The post-processing method based on SVD consistently achieves better performance than the wavelet transform. However, MSE captures only pointwise numerical accuracy and may not fully reflect structural or visual fidelity. As shown in the last two rows of Figure 8, Figure A.1, and Figure A.2, which correspond to modularity changes and TV ratios respectively, although SVD yields lower MSE across privacy budgets, wavelet-based post-processing better preserves structural properties and visual smoothness at larger  $\epsilon$ . This divergence indicates that numerical, structural, and visual fidelity are not necessarily aligned.

For SBC visualization, pre-processing methods are not applicable. We apply the median filter technique as the post-processing method, resulting in a noticeable improvement in visualization quality (see Figure 6). These results suggest that different visualization types may require distinct pre- and/or post-processing methods.

### 5.3. Workflow Effectiveness

For connectogram visualization, the proposed workflow yields results that closely resemble those obtained without privacy constraints (see Figure 9). In addition, the detected functional abnormalities in SZ are primarily concentrated in the subcortical (SC), auditory (AU), sensorimotor (SM), visual (VI), and cerebellar (CB) domains [Du et al. \(2020\)](#). Given that identifying these brain domains and regions is more critical than precisely estimating individual connectivity values, we allocate a larger portion of the privacy budget to the domain and region identification stage of the workflow. Similar outcomes are observed in SBC visualization, where the privacy-preserving SBC map closely aligns with the SBC map without privacy protection. For instance, Figure 10 illustrates strong connectivities between the selected seed PCC and its neighboring voxels.



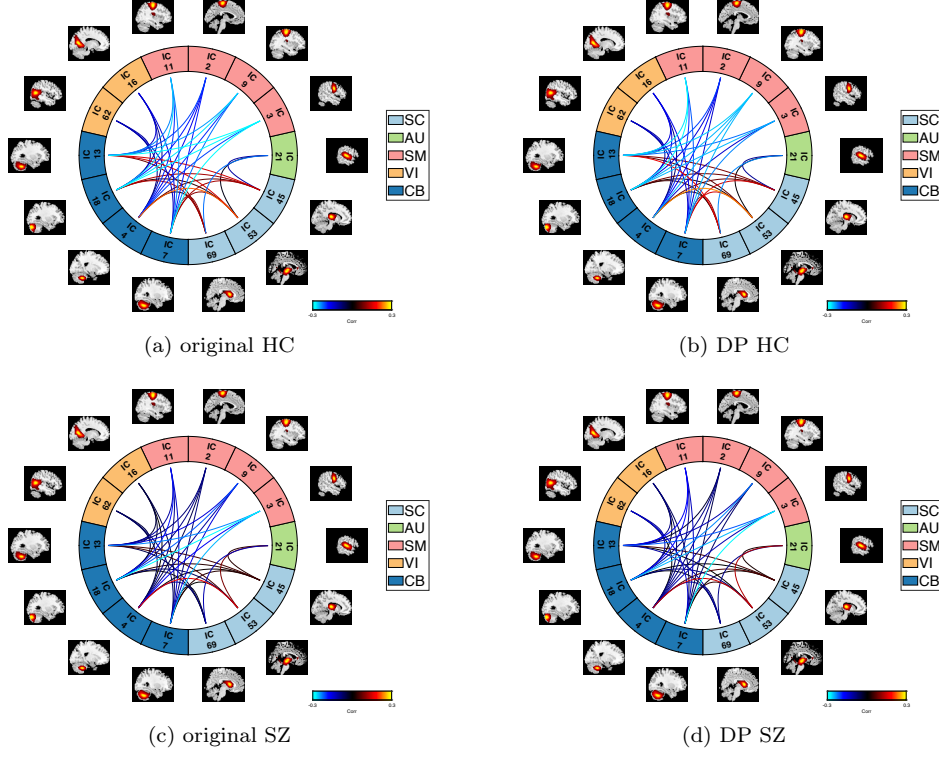


Figure 9: Privacy-preserving connectogram visualization is conducted with  $\epsilon = 2$ ,  $p_d = 5$ , and  $p_r = 14$ . The privacy budget is distributed as follows: 45% is assigned to identifying brain domains, 45% to selecting brain regions, and the remaining 10% to perturbing the connectivities. This allocation strategy prioritizes structural fidelity by assigning a larger portion of the privacy budget to domain and region identification, while allocating a smaller portion to connectivity values, which may introduce greater variability without substantially affecting the overall visual appearance of the connectogram. This design helps preserve group-level differences between HC and SZ under privacy constraints.

#### 5.4. Visual Utility

Evaluating the quality of privacy-preserving visualizations remains an open challenge due to the lack of a standard quantitative metric. Prior work has highlighted the difficulty of formally defining “visual utility” (Zhang et al., 2016). While numerical errors are commonly measured using  $\ell_1$  or  $\ell_2$  norms, such metrics may not necessarily reflect how well structural or perceptual patterns are preserved. Moreover, norm-based metrics are not always appropriate for evaluating visualization quality. For connectogram visualization, errors in hypothesis testing, such as incorrectly identifying non-



significant domains or failing to detect truly significant regions, may provide a more meaningful way to evaluate, since the primary concern is to identify domains or regions that differ significantly across groups. When it comes to SBC visualization, identifying quantitative methods to assess visual utility becomes even more challenging.

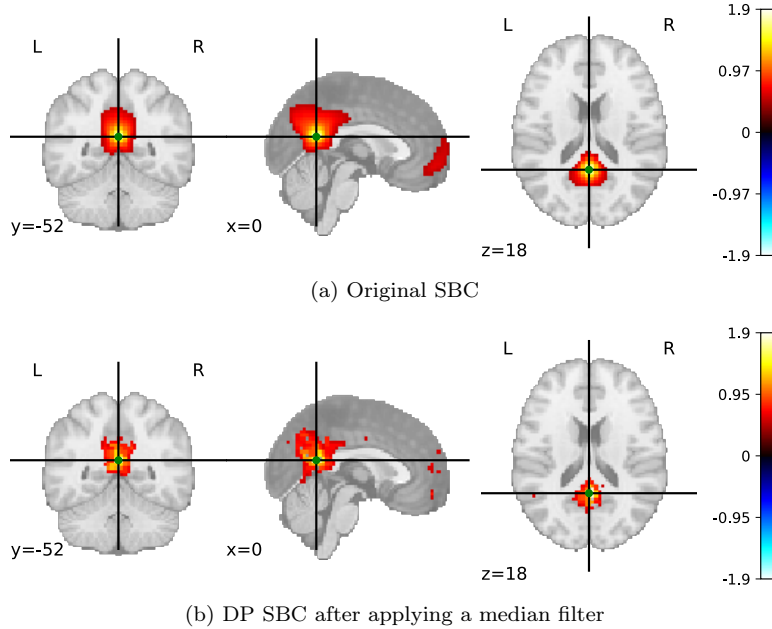


Figure 10: SBC maps on the NITRC dataset using the Posterior Cingulate Cortex (PCC) as the seed (indicated by the green dot), with parameters  $N = 1082$ ,  $\epsilon = 5$ , and  $\delta = N^{-3/2}$ . We allocate 50% of the privacy budget to identify significant correlations and 50% to protect the  $z$ -scores.

We surveyed neuroimaging researchers (at various levels of experience) through a convenience sample of researchers interested in functional network connectivity. We designed a survey consisting of five sections and collected 22 responses, with 18 participants consenting to the release of their responses. The full questionnaire, participant responses, and a summary of survey results are provided in the supplementary material. Our findings from this initial survey are as follows:

- Algorithm comparison: Participants found **Approx**-based methods, especially those with clipping and SVD, most similar to the original. In contrast, wavelet-based versions, although achieving lower MSE, were

frequently rated as more visually distorted. This suggests that quantitative metrics like MSE alone may not fully capture perceptual quality.

- Privacy-utility tradeoff: As the privacy budget  $\epsilon$  increased, visual quality tended to improve, showing clearer structures and fewer distortions.
- Consistency under the same privacy level: Most participants felt that visualizations generated using the same algorithm and privacy level were generally similar, despite differences in noise realizations.
- Connectogram workflow evaluation: Participants observed that allocating more privacy budget to brain region (node) identification improved perceived visualization quality. Conversely, reducing the budget for nodes increased visual distortions. This effect was not primarily caused by node misidentification itself but rather because node errors propagated to edges, resulting in larger and more noticeable changes in the central visual area.
- SBC workflow evaluation: Participants generally considered the privacy-preserving visualization usable and effective at preserving core connectivity patterns. However, some noted reduced smoothness and missing functional connections, suggesting that a higher privacy budget may be needed to better retain critical connections.

We believe this initial feedback provides some guidance for future research directions. In particular, we believe that for privacy-preserving visualization to become part of the regular workflow in future software systems for neuroimaging research, it is important to assess the visual utility as judged by the intended audience for the visualization.

## 6. Future Directions

Data visualizations are intended to support qualitative interpretation of quantitative data. We discussed the risks of sensitive information leakage in visualizations, highlighting the growing importance of privacy protection in this area. Current research has identified various privacy attacks across fields such as machine learning (Rigaki and Garcia, 2023). Adversaries may seek to access sensitive information, including the training dataset, model

parameters, hyperparameters, and architecture, etc. Inspired by these concerns, we believe that privacy attacks in visualizations are also an area worth exploring.

When working with sensitive data, the dichotomy between qualitative interpretation and quantitative precision creates opportunities for designing privacy-preserving mechanisms. In this study, we explored several approaches to providing privacy for visualizations related to functional network connectivity. We view this as a *starting point* for considering privacy in data visualization within neuroimaging and human health research more broadly. In such settings, individual data holders often have limited sample sizes. However, in federated learning systems, the larger aggregated sample size reduces the amount of noise required, which in turn diminishes the performance gap between different methods.

We did not explore the full gamut of post-processing methods or transform-based privacy algorithms. Additional directions for improvement include exploring alternative processing methods, such as Fourier transforms or bit plane slicing. Another possibility is to operate in transformed domains where the data admits sparse representations, enabling privacy-preserving protection of the coefficients prior to reconstruction via inverse transforms. Beyond these methods, visualization quality may also be influenced by design choices, including the selection of color maps, the number of discrete color levels  $|\mathcal{C}|$ , and the quantization strategy.

Many visualizations seek to represent the uncertainty in estimates or observations. Uncertainty visualization has long been recognized as a critical component for scientific communication, with numerous studies highlighting its importance (MacEachren, 1992; MacEachren et al., 2005; Potter et al., 2010). Visualizing uncertainty and providing privacy can potentially present synergies: designing effective algorithms is an interesting avenue for future work. A starting point would be to develop privacy-preserving displays for error bars and confidence intervals. When privacy is included, we might find that other visual representations of uncertainty are more visually salient.

## Ethics Statement

This project was designated as not human subjects research because it involved secondary analyses of publicly available, de-identified data and therefore did not require IRB approval. As indicated in the original publications, the original datasets used in this study were collected under the oversight of

the appropriate IRBs or Ethics Committees, and written informed consent was obtained from all participants prior to data collection.

## **Data and Code Availability**

This study utilizes publicly available datasets accessible directly from their respective repositories; signing a Data Use Agreement (DUA) may be required. The code is available from the corresponding author upon reasonable request.

## **Conflicts of Interest**

The authors declare no conflicts of interest that may bias or could be perceived to bias this work.

## **Acknowledgments**

The work of authors was supported in part by the U.S. National Institutes of Health under awards 2R01DA040487, R01DA049238, and R01MH121246, as well as by the National Science Foundation under award 2112455. The work of Sarwate was also supported by the National Science Foundation under award CNS-2148104 and is supported in part by funds from federal agency and industry partners as specified in the Resilient & Intelligent NextG Systems (RINGS) program.

The authors wish to thank the FBIRN, BLSA, and ADNI teams. In addition, the authors acknowledge the Neuro Bureau, the ADHD 200 consortium, and Virginia Tech's ARC.

The BLSA is supported by the National Institute on Aging Intramural Research Program, NIH. This research was supported in part by the Intramural Research Program of the NIH. The contributions of the NIH author(s) are considered Works of the United States Government. The findings and conclusions presented in this paper are those of the author(s) and do not necessarily reflect the views of the NIH or the U.S. Department of Health and Human Services.

## **Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process**

During the preparation of this work the authors used ChatGPT in order to improve the language quality. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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## Appendix

The appendix contains additional details and figures omitted from the main manuscript.

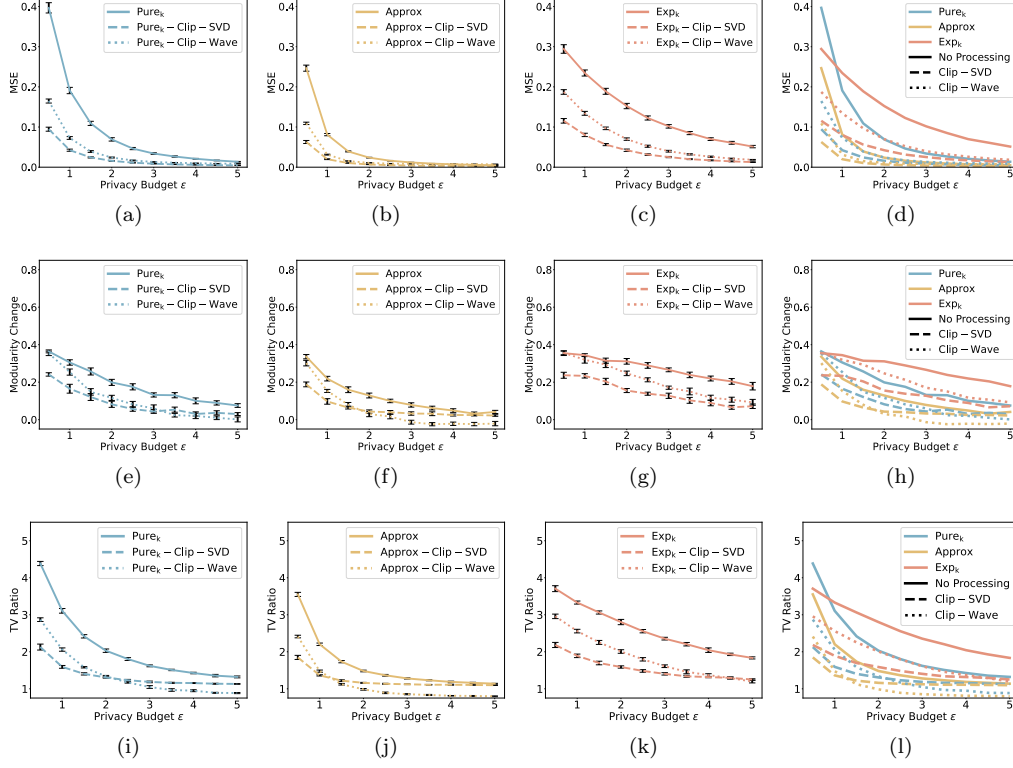


Figure A.1: Quantitative evaluation of privacy-preserving FNC visualization on the BLSA dataset. The first row reports MSE between the original and private average FNC matrices. The second row shows modularity change, defined as  $Q_{\text{original}} - Q_{\text{DP}}$ , where  $Q$  denotes signed Louvain modularity. The third row presents the TV ratio, computed as  $\text{TV}_{\text{DP}}/\text{TV}_{\text{original}}$ . Columns correspond to (a)(e)(i)  $\text{Pure}_k$ -, (b)(f)(j)  $\text{Approx}$ -, (c)(g)(k)  $\text{Exp}_k$ -related methods, and (d)(h)(l) all methods. Each configuration is repeated 50 times, and error bars are provided.

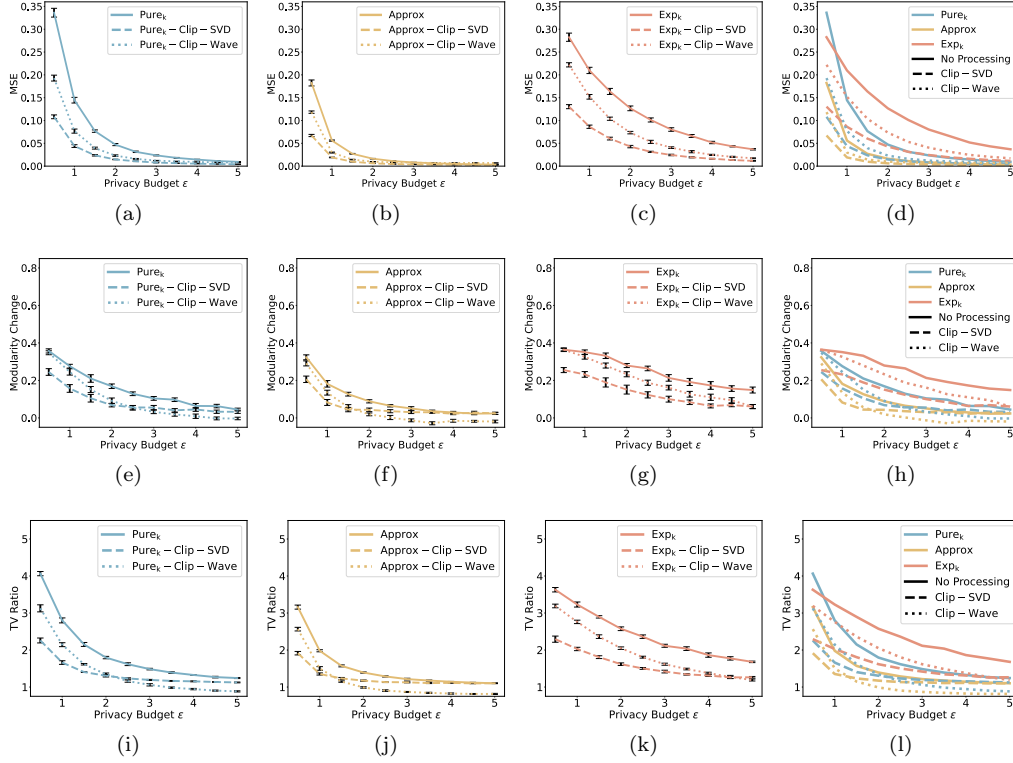


Figure A.2: Quantitative evaluation of privacy-preserving FNC visualization on the ADNI dataset. The first row reports MSE between the original and private average FNC matrices. The second row shows modularity change, defined as  $Q_{\text{original}} - Q_{\text{DP}}$ , where  $Q$  denotes signed Louvain modularity. The third row presents the TV ratio, computed as  $\text{TV}_{\text{DP}}/\text{TV}_{\text{original}}$ . Columns correspond to (a)(e)(i) Pure<sub>k</sub>-, (b)(f)(j) Approx-, (c)(g)(k) Exp<sub>k</sub>-related methods, and (d)(h)(l) all methods. Each configuration is repeated 50 times, and error bars are provided.

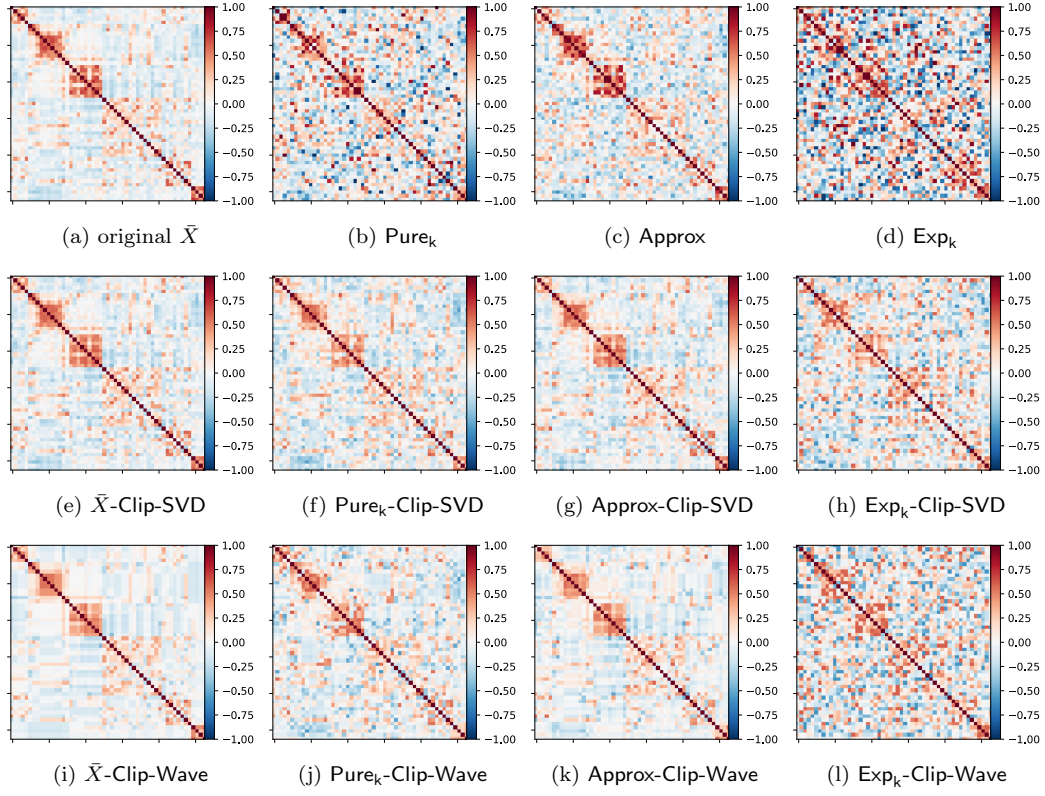


Figure A.3: Visualization of the BLSA dataset with  $N = 891$ ,  $\epsilon = 2$ ,  $\delta = N^{-3/2}$ , clipping bound  $b = 0.7$ , number of components used in SVD set to 10, threshold used in the wavelet transform set to 0.2, and number of colors set to 256.

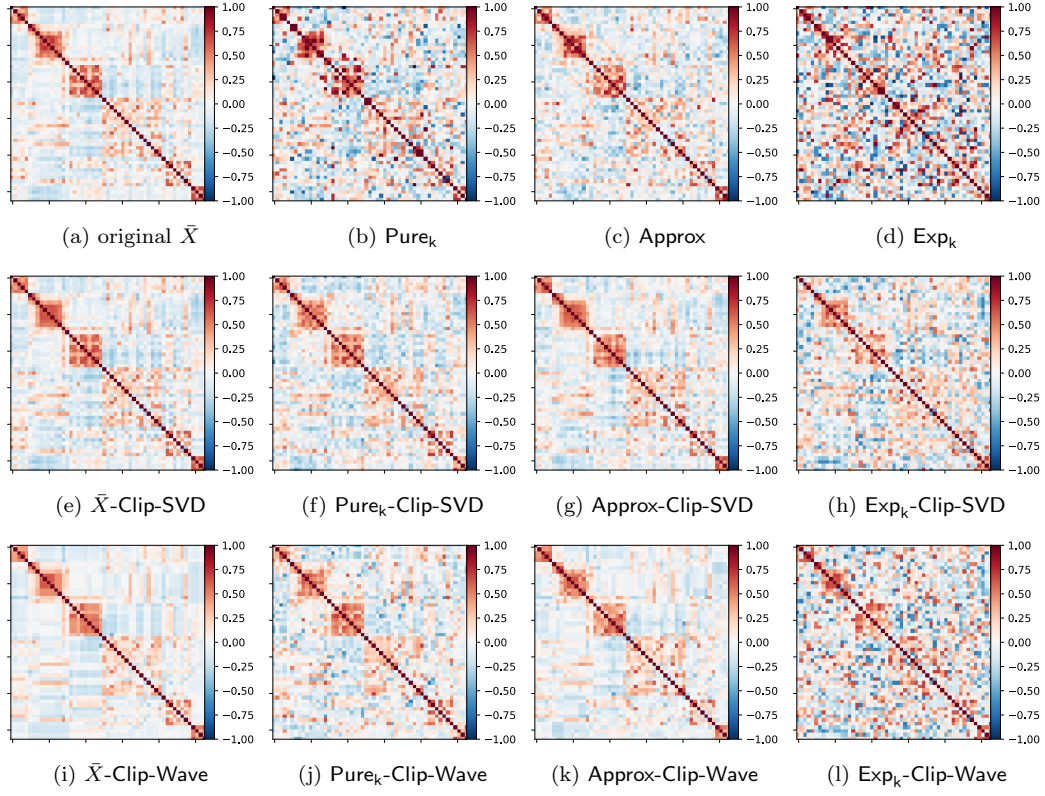


Figure A.4: Visualization of the ADNI dataset with  $N = 1091$ ,  $\epsilon = 2$ ,  $\delta = N^{-3/2}$ , clipping bound  $b = 0.75$ , number of components used in SVD set to 10, threshold used in the wavelet transform set to 0.2, and number of colors set to 256.